USER STORIES AND TASKS SPECIFICATION

Project A- Informix Monitoring Dashboard:

User story

US-01: View database health and resource utilization

As a database administrator, I want to view CPU, memory, and I/O performance metrics in a centralized Informix dashboard so that I can monitor overall database health, identify potential resource bottlenecks, detect performance degradation early, and make informed decisions regarding capacity planning and system optimization before service levels are impacted.

Tasks:

- T1. Identify the Informix health and performance metrics to monitor.
- T2. Develop the mechanism for collecting the required metrics.
- T3. Process and present the metrics in a web dashboard.
- T4. Organize the dashboard into clear monitoring sections.
- T5. Validate the displayed values against the available database or operating system measurements.

Acceptance criteria:

- The dashboard displays the implemented CPU, memory, and I/O metrics.
- Available performance measurements are presented in a readable format.
- Displayed values can be validated against their underlying data sources.
- Missing or unavailable measurements are not misleadingly presented as valid readings.

US-02: Investigate database performance issues

As a DBA, I want to review key database performance indicators through a centralized dashboard so that I can quickly identify abnormal trends, investigate potential database performance issues, assess their impact on system health and availability, and prioritize further troubleshooting and remediation activities based on the severity of the findings.

Tasks:

- T1. Identify indicators that help reveal performance degradation.
- T2. Implement or configure the relevant dashboard visualizations.
- T3. Review available measurements for unusual values or trends.
- T4. Validate suspected issues using appropriate database or operating system tools.

Acceptance criteria

- Relevant performance indicators can be accessed from the dashboard.
- The displayed data supports investigation of potential performance issues.
- Findings can be checked against the underlying measurements.

Project B- DB Lab Performance

User story

US-03: Establish a realistic transaction workload

As a DBA, I want to populate a banking database with 2 million representative transactions so that I can evaluate database behavior under a substantial transaction workload, measure performance and resource utilization, identify potential bottlenecks, validate system stability, and assess the database's ability to support high-volume operational demands.

Tasks:

- T1. Design and create the required banking database schema.
- T2. Prepare the transaction data generation or loading process.
- T3. Load the 2 million transactions into the database.
- T4. Validate the loaded data and transaction count.
- T5. Record the test environment and workload assumptions.

Acceptance criteria:

- The banking database schema was created successfully.
- The database contains 2 million loaded transaction records, as verified by an appropriate count.
- The data can be queried for subsequent performance testing.
- The test environment and relevant workload details are documented.

US-04- Identify and optimize slow queries

As a DBA, I want to identify slow-running queries and optimize their execution by implementing appropriate indexing strategies so that the banking database can process queries more efficiently, reduce response times, improve transaction throughput, and maintain acceptable performance under increasing workload demands.

Tasks:

- T1. Identify queries with poor execution performance.
- T2. Measure baseline execution times or other relevant performance indicators.
- T3. Inspect query plans and identify indexing opportunities.
- T4. Create or modify appropriate indexes.
- T5. Rerun the queries and compare the results with the baseline.
- T6. Document the performance findings and any trade-offs.

Acceptance criteria:

- The selected slow queries are identified and documented.
- Baseline measurements are recorded where available.
- Index changes are supported by analysis of the relevant queries.
- The queries are retested after the changes.
- Any performance improvement claimed is supported by measured results.

US-05- Back up the database and verify recovery

As a DBA, I want to back up the banking database and verify that the backup can be successfully restored so that I can confirm the integrity and reliability of the backup process, ensure critical banking data can be recovered in the event of a failure, and minimize the risk of data loss and service disruption.

Tasks:

- T1. Identify the appropriate backup method for the database platform.
- T2. Execute the backup procedure.
- T3. Check the backup results and relevant logs.
- T4. Restore the backup in an appropriate test environment.
- T5. Validate the restored database using suitable checks.
- T6. Document the recovery procedure and test outcomes.

Acceptance criteria:

- The backup process completes with a verifiable outcome.
- The backup can be restored successfully in the test environment.
- Appropriate checks confirm that the restored database is usable and contains the expected data.
- The procedure and any issues encountered are documented.

US-06-Monitor database health

As a DBA, I want an automated health check to evaluate selected database health indicators on a scheduled basis so that I can proactively identify potential performance, availability, and capacity issues, investigate anomalies promptly, and take corrective action before they impact database services or business operations.

Tasks:

- T1. Define the database health checks to perform.
- T2. Develop a script to execute the checks.
- T3. Implement clear output for successful and unsuccessful checks.
- T4. Test the script against the expected conditions.
- T5. Document how to run the check and interpret its output.

Acceptance criteria:

- The health check evaluates the indicators implemented in the script.
- Results are reported in a clear, interpretable format.
- Failure conditions are distinguishable from successful checks.
- The script's behavior is tested and documented.

Project C – Live banking Fraud Detection

User Story

US-07-Evaluate incoming card transactions

As a fraud detection system, I want to evaluate incoming card transactions against configured fraud detection rules in real time so that potentially suspicious activity can be identified as transactions arrive, enabling timely investigation, risk mitigation, and prevention of fraudulent transactions before financial losses occur.

Tasks:

- T1. Identify the incoming transaction fields required by the rules.
- T2. Define or configure the agreed fraud detection rules.
- T3. Implement the transaction evaluation logic.
- T4. Test the rules using suitable transaction scenarios.
- T5. Verify that the evaluation produces the expected outcomes.

Acceptance criteria:

- Incoming transactions are evaluated against the configured rules.
- Test transactions produce results consistent with the expected rule behavior.
- Transactions that match configured suspicious-activity rules are identified for further review.
- The system handles invalid or incomplete input according to its defined behavior.

US-08- Display suspicious activity to analysts

As a fraud analyst, I want suspicious transaction alerts to appear on a dashboard as soon as they are generated so that I can promptly investigate potentially fraudulent activity, assess the associated risk, take appropriate action to prevent financial losses, and ensure that genuine customer transactions are not unnecessarily disrupted.

Tasks:

- T1. Define the information required for an alert.
- T2. Connect the fraud evaluation output to the dashboard.
- T3. Implement the display of generated alerts.
- T4. Test the flow from transaction receipt to alert presentation.
- T5. Verify the accuracy and timeliness of the displayed information.

Acceptance criteria:

- Transactions identified by the rules engine generate the expected alerts.
- Generated alerts appear on the analyst's dashboard.
- Alert information corresponds to the relevant transaction and detection results.
- The transaction-to-alert flow is tested under the supported operating conditions.

US-09- Review and classify fraud alerts

As a fraud analyst, I want to review suspicious transaction alerts and confirm them as fraud or dismiss them as false positives so that potentially fraudulent activity can be accurately classified, appropriate actions can be taken to mitigate risk, and legitimate customer transactions can proceed without unnecessary disruption.

Tasks:

- T1. Implement or configure the alert review interface.
- T2. Provide the supported fraud-confirmation and false-positive actions.
- T3. Record the review outcome against the relevant alert.
- T4. Test both classification outcomes.
- T5. Verify that the resulting status is displayed and retained as intended.

Acceptance criteria:

- An authorized analyst can perform the supported review actions.
- Each action produces the correct alert status.
- The outcome remains associated with the correct alert.
- The system handles unsuccessful or invalid review actions appropriately.